

SmolVLA for Language-Conditioned Driving: Matched-Input and Closed-Loop Evaluation

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Abstract—Compact vision–language–action (VLA) models enable language-conditioned robot control, but instruction use after adaptation to a vehicle embodiment remains insufficiently characterized. We fine-tune SmolVLA for simulated closed-loop control on 406 language-tagged demonstrations comprising 93,733 samples and 388 task strings. The policy maps front RGB, English instruction, speed, yaw rate, and a constant legacy state slot to a 30-step incremental SE(2) chunk. A 2^3 matched-input intervention varies the template, the tested *inner/outer-to-innermost/outermost* replacement, and speed wording across 20 fixed observations. Base wording gives 0.690-m median raw action-path separation versus a 0.125-m lane-1 same-instruction reference. Template- and speed-only changes retain 0.826 and 0.672 m, whereas the tested lane replacement reduces separation to 0.093 m; the pattern persists after an offline bridge transform. In 32 wall-clock rollouts balanced across two starts and both instruction orders, Base different-instruction pairs yield 7.376-m median trajectory separation versus a 0.177-m repeated-instruction reference. Every post-15-m resampled point in all 16 Base rollouts is closer to the requested lane, whereas all eight inner requests under the test wording are closer to the outer reference. The results establish a measured language-to-motion path and localize the observed loss to this tested replacement rather than sentence novelty in general.

Index Terms—vision–language–action models, SmolVLA, language-conditioned driving, action chunking, closed-loop evaluation, ROS 2

I. INTRODUCTION

Vision–language–action models promise a common interface between natural language, visual perception, and continuous robot control. OpenVLA showed that an open pretrained VLA can be adapted to multiple manipulation platforms and tasks [1], while SmolVLA reduced the policy scale to roughly 450M parameters and retained chunked continuous actions [2]. The smaller model is attractive for robot and vehicle prototypes, but adapting it to a new embodiment raises a basic empirical question: does the fine-tuned policy use the instruction to choose motion, or does it mostly reproduce the visually likely route?

Vehicle driving makes that distinction observable. Two lane instructions can share the same initial camera view and vehicle state yet require different future paths. A policy that ignores language may still drive smoothly, so completion or lane keeping alone cannot establish language conditioning. Con-

versely, a difference between two stochastic policy samples is not by itself evidence that changing the instruction caused the difference. Evaluation must therefore hold the visual and state observations fixed, compare the resulting action chunks against a disjoint-seed same-instruction reference, and determine whether the separation persists after the actions enter a closed loop.

We study a SmolVLA checkpoint fine-tuned on language-tagged demonstrations for a simulated vehicle-like platform. The policy receives one front image, an English sentence, speed, yaw rate, and a constant legacy state slot, and predicts 30 incremental ego-motion actions. It is deployed through a ROS 2 bridge in Gazebo, where the bridge deterministically maps the checkpoint output to motion commands. Rather than claim broad natural-language understanding, we compare Base wording assembled from observed components with a symmetric test wording pair assembled from phrase-bank components absent from the vehicle fine-tuning inventory. None of the 16 complete evaluation sentences occurs verbatim in the recovered final task inventory. We vary the three wording components factorially to identify which tested change is associated with lost lane response.

The contributions are threefold:

- 1) a compact-VLA adaptation pipeline for language-conditioned vehicle control, from validated demonstrations to 30-step incremental SE(2) chunks and ROS 2–Gazebo execution;
- 2) a matched-observation 2^3 intervention that separates template, the tested *inner/outer-to-innermost/outermost* replacement, and speed-term effects and checks both raw paths and an offline bridge-transformed sensitivity path; and
- 3) 32 balanced closed-loop rollouts showing that the baseline language distinction survives vehicle execution, together with the closed-loop failure observed for that tested replacement.

II. RELATED WORK

OpenVLA established open downstream VLA adaptation, while SmolVLA combines a compact backbone with flow-matching action chunks [1], [2]. ROS2SmolVLA integrates

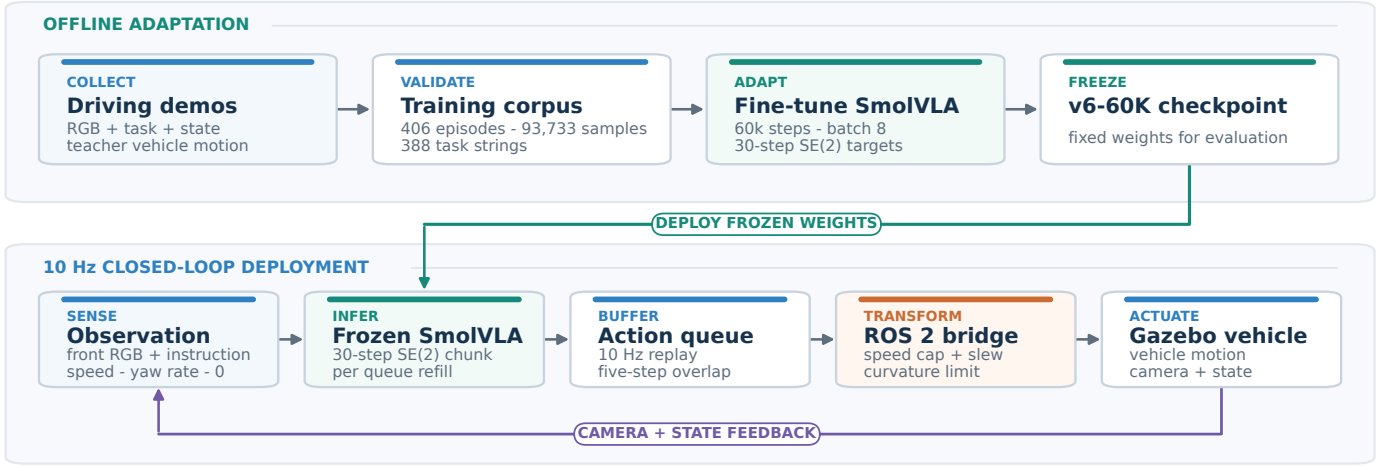


Fig. 1. Training and deployment pipeline. Frozen SmolVLA maps RGB, language, and state to 30-step SE(2) chunks replayed at 10 Hz in ROS 2–Gazebo.

SmolVLA with an industrial manipulator through ROS 2 [3]; our embodiment and evaluation instead concern planar vehicle control and the measured effect of driving instructions.

Within our laboratory’s scaled-vehicle lineage, Oh *et al.* developed a competition-based autonomous-driving education framework [4], and Hong *et al.* compared rule-based and learned end-to-end vehicle control [5]. Suh *et al.* evaluated a modular detector–VLM driving stack [6] and later a latency-aware slow/fast safety-arbitration architecture [7]. These works establish the platform, learning, and VLA lineage for the present study. We reuse that lineage’s simulated track and its fixed inner/outer lane identities, but each episode instruction here names a lane, a speed term, and often a destination or stop target rather than a single lane-change command.

SimLingo studies language-aligned closed-loop driving, and OpenDriveVLA predicts trajectories from perception, state, and driver commands [8], [9]. LinkVLA directly trains language–action alignment [10], while ICR-Drive isolates instruction effects under controlled text changes [11]. Manipulation benchmarks report the same class of failure: counterfactual instruction assignment under visually plausible layouts exposes vision shortcuts [12], and synonym-level paraphrase alone degrades task success [13]. We therefore do not claim the failure mode itself as new. In contrast to these evaluations, we freeze the exact RGB/state input for a factorized action-chunk intervention on a vehicle policy and then test whether the measured distinction persists through a ROS 2–Gazebo closed-loop executor.

III. METHODOLOGY

Figure 1 summarizes the full system, and Table I lists the training and deployment configuration. The learned component is the fine-tuned SmolVLA policy; the same frozen checkpoint is used in every reported evaluation.

A. Demonstration Corpus

A teacher controller generated ring-driving motion while an episode-level English instruction specified lane and speed, often together with a landmark destination or stop task.

Each raw episode contained a front-camera stream, the language label, vehicle state, teacher control, and global pose at 30 Hz. Validation rejected incomplete or geometrically invalid episodes, and retained records were resampled into a 10-Hz LeRobot-style representation. The final v6 LeRobot metadata records 406 episodes, 93,733 action rows, and 388 distinct task strings, which compose lane, speed, and destination slots over paraphrased surface forms across six named track targets and a lap form. That composition is a premise of the later test: lane is one slot among three, so a lane-dependent response cannot come from matching two fixed strings, and the template and speed slots supply the other two wording factors. We recovered the complete task inventory from the training artifact and stored its sorted hash with the evaluation records.

The checkpoint’s three-value state is

$$\mathbf{s}_t = [v_t, \dot{\psi}_t, 0], \quad (1)$$

where v_t and $\dot{\psi}_t$ denote vehicle speed and yaw rate. The third, legacy state slot is identically zero in all 93,733 training rows and is forced to zero at evaluation; it is not treated as a measured steering signal. Global position and map coordinates are excluded from policy input. A training target is the ego-frame displacement over one 0.1-s step, $\mathbf{a}_i = [\Delta x_i, \Delta y_i, \Delta \psi_i]$. Language is therefore the only direct specification of which lane the policy should follow when the initial visual and state observations are identical.

B. Policy and Fine-Tuning

Starting from the compact SmolVLA release, we fine-tune a policy that predicts

$$\mathbf{A}_t = \pi_\theta(I_t, \ell, \mathbf{s}_t), \quad \mathbf{A}_t \in \mathbb{R}^{30 \times 3}, \quad (2)$$

where I_t is the front RGB image and ℓ is the English task. Training runs for 60,000 optimization steps with batch size 8, learning rate 10^{-4} , automatic mixed precision, seed 1000, and no image augmentation. This corresponds to approximately 5.12 passes over the 93,733-row corpus. The resulting v6-60K checkpoint is frozen for all experiments.

The checkpoint retains SmolVLA’s SmolVLM2-500M-Video-Instruct backbone, cross-attention action conditioning, and ten flow-matching denoising steps. We update the full policy with AdamW, a 1,000-step warmup, and cosine decay; all settings are held fixed across language conditions.

The predicted increments are composed into an ego-frame path for analysis:

$$\mathbf{p}_0 = \mathbf{0}, \quad \mathbf{p}_{i+1} = \mathbf{p}_i \oplus \mathbf{a}_i, \quad i = 0, \dots, 29, \quad (3)$$

where $\oplus$ applies the translation after rotating it by accumulated yaw. This is a diagnostic path composed from the *raw* policy output, including Δy ; it is not a simulated vehicle trajectory. Deployed replay uses Δx and $\Delta \psi$, as specified below, and closed-loop evaluation measures the resulting vehicle motion.

TABLE I
TRAINING AND DEPLOYMENT CONFIGURATION

Item	Configuration
Policy	SmolVLA, approx. 450M parameters; frozen v6-60K checkpoint
Corpus	406 episodes; 93,733 rows at 10 Hz; 388 task strings
Observation	Front RGB; English task; speed, yaw rate, constant-zero legacy slot
Action	$30 \times [\Delta x, \Delta y, \Delta \psi]$ at 0.1-s intervals
Fine-tuning	60K steps; batch 8; LR 10^{-4} ; AMP; seed 1000; no image augmentation
Queue	10 Hz; five-step overlap; refill at 30% remaining; direct replay
Command limits	2.25-m/s speed; 0.08-m/s/tick slew; 0.239-m^{-1} curvature
Evaluation	One ring; two matched starts; 30-s wall-clock runs; policy seed 0

C. ROS 2–Gazebo Deployment

The ROS 2 deployment [14] sends the image, state, and latched task to a policy server and receives a 30-step action chunk. A bridge maintains a queue with five-step overlap and requests a new chunk when 30% remains. At each 10-Hz tick, direct replay obtains the raw speed and curvature,

$$v_i^0 = \frac{\Delta x_i}{0.1}, \quad \kappa_i^0 = \frac{\Delta \psi_i}{\Delta x_i} \quad (|\Delta x_i| > 10^{-4} \text{ m}), \quad (4)$$

then caps speed at 2.25 m/s, limits its per-tick change to 0.08 m/s, clips κ_i^0 to $\pm 0.239 \text{ m}^{-1}$, and publishes $\omega_i = \kappa_i v_i$. When $|\Delta x_i| \leq 10^{-4} \text{ m}$, the raw yaw rate is retained instead. The lateral prediction Δy_i is not sent to the executor. Gazebo closes the camera and state loop around a vehicle-shaped model whose body motion is implemented by a differential-drive plugin [15]. Front steering joints visualize the corresponding curvature but do not define the body dynamics. No map, goal vector, or global pose is sent to the policy.

The server senses one front image, takes its camera key from the checkpoint, and recovers the actual three-value state width from the saved normalizer. Inherited unused camera slots and the base configuration’s placeholder six-value state shape are ignored.

For evaluation, the simulator is paused after each matched commanded-pose reset, the instruction is published, and execution begins only after the first action chunk arrives. This prefill protocol prevents startup delay from being counted as a controller watchdog failure; it does not change the predicted actions or the subsequent closed-loop control. A rollout is accepted only if the commanded position is verified within 5 cm, the trajectory contains at least five retained points, and both watchdog activations and action underruns remain zero.

D. Evaluation Protocol

1) *Matched-observation action test*: We sample 20 validated lane-counterfactual groups from three collection sessions in the v6 training lineage and select 10-Hz row index 5 (nominally 0.5 s after the common start). For each group, the exact same RGB array and state vector are supplied to both sides; only the instruction differs. A 2^3 design changes three binary wording components: template T (“Start driving”/“Roam the track”), lane replacement L (“inner, outer”/“innermost, outermost”), and speed term S (“normal”/“regular”). This gives Base, T , L , S , $T+L$, $T+S$, $L+S$, and $T+L+S$ cells with unchanged lane semantics.

None of the 16 complete evaluation sentences appears verbatim in the recovered 388-task training inventory. The Base components occur in that inventory, whereas all alternate T , L , and S components occur zero times and belong to a phrase bank committed before v6 training. Thus the design localizes the response within this fixed wording set; the visual observations themselves are not held out.

SmolVLA sampling is stochastic. We average four predicted chunks for each instruction using seeds 0–3. For context, the same image, state, and lane-1 sentence are sampled again with disjoint seeds 4–7. This is an asymmetric lane-1 same-instruction reference, not an upper bound on stochastic variation for both instructions. Let $P_{xy}(\ell) = \{\mathbf{p}_{i,xy}(\ell)\}_{i=1}^{30}$ contain the translation components of the raw composed poses from (3). Instruction separation is

$$D_{\text{chunk}}(\ell_1, \ell_2) = \sqrt{\frac{1}{30} \sum_{i=1}^{30} \|\mathbf{p}_{i,xy}(\ell_1) - \mathbf{p}_{i,xy}(\ell_2)\|_2^2}. \quad (5)$$

For each observation we also report $R = D_{\text{chunk}}/D_{\text{ref},1}$, where $D_{\text{ref},1}$ applies (5) to the disjoint-seed lane-1 chunks; $R > 1$ exceeds this scoped same-instruction reference. Factor effects are saturated contrasts of $\log_2(D_{\text{chunk}}/1 \text{ m})$, computed within each observation and summarized with a 50,000-resample paired bootstrap. The intervals are exploratory point-wise intervals over this fixed sample.

The saved four-seed mean chunks also support a zero-initial-command bridge-transform sensitivity check. Although the observation speed spans 0.562–2.181 m/s, this diagnostic initializes the *previous commanded* speed to zero, drops Δy , applies the recorded speed cap and slew together with the source-defined curvature clip in (4), integrates the resulting unicycle commands, and recomputes separation. It does not

emulate queue refill, splicing, or Gazebo; those effects are evaluated only by the closed-loop runs.

2) *Paired closed-loop test*: The closed loop evaluates only the two factorial endpoints, Base and $T+L+S$; localization of the individual factors is action-level only. We use inner- and outer-lane starts at the same longitudinal track index. For each wording regime and start, one invocation uses inner then outer (AB) and a second uses outer then inner (BA). Each invocation records a same-instruction pair (A, A) followed by a different-instruction pair (A, B). Thus both same-inner and same-outer references and both instruction orders are observed at each start: 2 regimes $\times$ 2 starts $\times$ 2 orders $\times$ 4 physical rollouts gives 32 rollouts. Each lasts 30 s wall clock with policy seed 0.

The spatially thinned trajectories are truncated to their common traveled length, and their discrete Fréchet distance is reported as D_{shape} . This metric measures path separation while respecting trajectory order. The shared prefix is the traveled distance before the resampled paths first differ by more than 0.30 m. Both are behavioral differentiation metrics; they do not by themselves establish correct lane occupancy or safe driving.

To add a directional check, we discard the first 15 m of each trajectory so the common start and lane transition do not dominate the result. For every remaining point $\mathbf{x}$, we compute its Euclidean distance to the two configured lane-center polylines, R_1 and R_2 . For an instruction requesting lane ℓ , we report the per-run median and 95th percentile of

$$d_\ell(\mathbf{x}) = \min_{\mathbf{r} \in R_\ell} \|\mathbf{x} - \mathbf{r}\|_2, \quad (6)$$

where the implementation projects onto line segments rather than only sampled vertices. We also report the fraction of analyzed points for which $d_\ell(\mathbf{x}) < d_{3-\ell}(\mathbf{x})$. This is a centerline-relative route check, not a lane-boundary or collision-safety metric.

3) *Reproducibility and validity controls*: Independent SHA-256 checks verify that the closed-loop and offline evaluations use the same checkpoint; the full digest is retained in the checkpoint manifest. The recovered training inventory and hash, 20 observation identifiers and state vectors, raw mean chunks, per-seed protocol, full closed-loop XY traces, exact strings, reset coordinates, unique rollout IDs, applied bridge contract, latency, watchdog count, and underrun fraction are retained in the project evaluation records. Plot and table values are regenerated without invoking the policy or simulator. No empty-language control is reported because an empty task disarms the deployed bridge.

IV. RESULTS

A. Instruction Conditioning of Action Chunks

Figure 2 plots all 20 observation-level language/reference ratios in each wording cell, while Table II reports exact summaries. Without L , median R is 5.14–6.15 and every cell exceeds the reference in 20/20 observations. With the tested replacement of *inner/outer* by *innermost/outermost* (L), median R falls to 0.42–0.52 with only 0–1/20 exceedances;

raw D_{chunk} similarly separates into 0.672–0.826 m without L and 0.062–0.093 m with L .

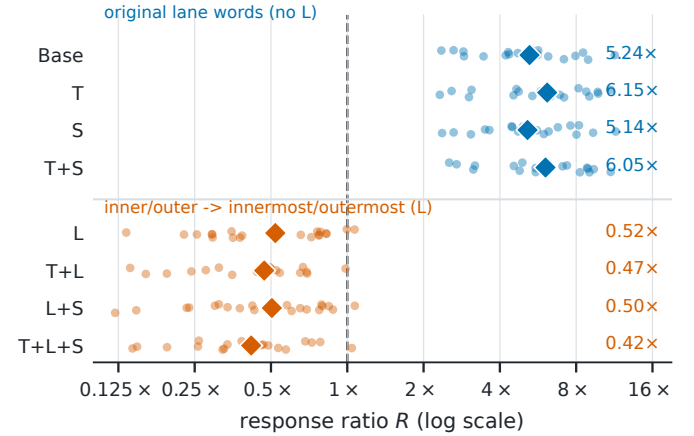


Fig. 2. Raw response ratio R across the eight wording cells. Small points are 20 matched observations and diamonds are medians; $R = 1$ (dashed) equals the disjoint-seed lane-1 same-instruction reference. T , L , and S denote template, the tested lane replacement, and speed terms.

TABLE II
2³ MATCHED-OBSERVATION RAW ACTION SEPARATION

Cell ^a	D_{chunk} Lang. (m)	Lane-1 ref. (m)	Median ratio ^b	Above ref.
Base	0.690	0.125	5.24	20/20
T	0.826	0.123	6.15	20/20
S	0.672	0.121	5.14	20/20
$T+S$	0.823	0.123	6.05	20/20
L	0.093	0.250	0.52	1/20
$T+L$	0.062	0.205	0.47	0/20
$L+S$	0.090	0.218	0.50	1/20
$T+L+S$	0.063	0.220	0.42	1/20

^a T : template; L : tested lane replacement; S : speed term.

^bMedian of per-observation language/reference ratios.

Separately, the raw- D_{chunk} factorial contrast gives an L effect of -3.118 on $\log_2(D_{\text{chunk}}/1 \text{ m})$, with a paired-bootstrap 95% interval of $[-3.469, -2.797]$, corresponding to a factorial geometric-mean multiplier of 0.115. On the multiplier scale, the T main effect and $T \times L$ interaction are 0.916 $[0.852, 0.984]$ and 0.828 $[0.769, 0.890]$, respectively. The log-scale intervals for S and the remaining interactions span zero. Thus L is the dominant, but not exclusive, negative factor in this fixed offline wording intervention.

The zero-initial-command bridge-transform check gives the same dominant factor: $L = -3.339$ $[-3.704, -3.009]$, a geometric-mean multiplier of 0.099. Its Base and $T+L+S$ medians are respectively 0.221 versus 0.031 m and 0.016 versus 0.037 m against their lane-1 references. All eight cells retain the same above/below-reference median classification under raw and transformed paths.

B. Closed-Loop Language Response

The closed-loop results in Fig. 3 and Table III provide a directional complement to the action-level test. Base word-

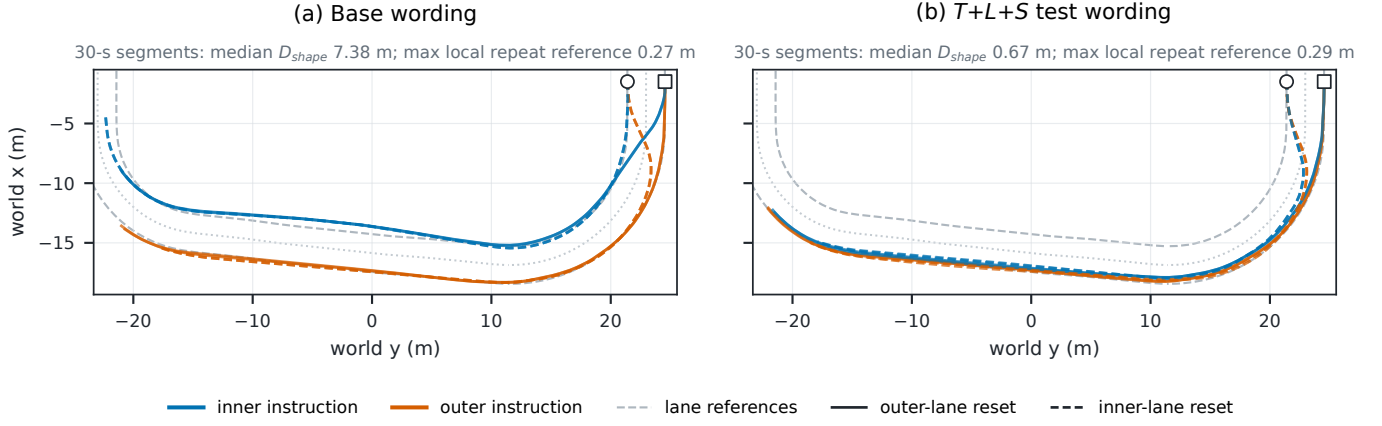


Fig. 3. Thirty-second wall-clock closed-loop segments from matched resets. Colors denote requested lanes and line styles denote reset lanes; gray curves are lane references. Base wording selects both routes, whereas the $T+L+S$ test wording collapses toward the outer route.

ing produces $D_{\text{shape}} = 7.376$ m across the four different-instruction pairs, compared with a 0.177-m median over the four same-instruction pairs. All four differences exceed the local repeated-instruction reference for their own reset, defined as the larger of the same-inner and same-outer values. The result holds at both starts and in both AB and BA orders; the first > 0.30 -m divergence occurs after a median 2.5 m.

TABLE III
PAIRED CLOSED-LOOP TRAJECTORY SEPARATION

Wording	Different D_{shape} median [range] (m)	Same D_{shape} median [range] (m)	Above local ref.	Shared prefix (m)
Base	7.376 [6.740, 7.860]	0.177 [0.167, 0.265]	4/4	2.5
$T+L+S$	0.666 [0.569, 0.889]	0.243 [0.173, 0.292]	4/4	10.3

Table IV checks whether the separation has the requested direction. After 15 m, all sampled points in all 16 Base-wording rollouts are closer to the requested reference. Across eight rollouts per target, the median of run-level median offsets is 0.339 m for inner and 0.086 m for outer; the corresponding median run-level 95th percentiles are 0.728 and 0.277 m.

TABLE IV
RUN-LEVEL CENTERLINE ADHERENCE AFTER 15 M (EIGHT RUNS/TARGET)

Wording	Request	Median of run medians (m)	Median of run P95s (m)	Median run fraction (%)
Base	Inner	0.339	0.728	100
Base	Outer	0.086	0.277	100
$T+L+S$	Inner	2.779	3.019	0
$T+L+S$	Outer	0.159	0.303	100

For $T+L+S$ wording, the different-instruction median is 0.666 m versus a 0.243-m same-instruction median, and the shared prefix increases to 10.3 m. Although all four differences exceed their local references, separation alone is not correctness. Every 0.2-m-resampled post-15-m point from all eight inner-request rollouts is closer to the *outer* reference, whereas

the corresponding points in all eight outer-request rollouts are closer to the outer reference. The same 0%/100% target split holds for 10-, 15-, and 20-m cutoffs and from both reset lanes. Base stay and change maneuvers each have a median closer-to-target fraction of 1.0; $T+L+S$ gives 0.5 for each because behavior depends on the outer-lane bias, not the requested maneuver.

All 32 rollouts satisfy the logged execution contract: watchdog hits and action underrun are zero, steering-state source is constant zero, and the applied speed cap is 2.25 m/s. First-chunk prefill is 0.239–0.299 s and the bridge’s running mean request round-trip time is 242.1–251.4 ms. The trajectories cover 52.74–58.98 m in 30 s wall clock, approximately 0.37–0.42 of the 141.97-m nominal centerline length rather than a full lap.

C. Discussion of Scope and Differentiation

The combined tests provide evidence at two interfaces. At the policy interface, Base inner/outer wording changes the 30-step motion prediction under identical perception and state. At the embodiment interface, the same distinction remains large after repeated perception, queue refill, and vehicle dynamics close the loop. Together, these results demonstrate the feasibility of language-conditioned control with the fine-tuned compact VLA in the evaluated simulator; no comparison against alternative policies is implied.

The result for the test wording is pair-specific: within this fixed factorial set, only combinations retaining *inner/outer* produce a robust lane-dependent action change. The tested *innermost/outermost* replacement does not establish a general failure on unseen lane terms. Since none of the 16 complete evaluation sentences occurs in the task inventory, exact-string novelty is insufficient evidence of language generalization. A vehicle VLA evaluation should factor wording components and measure each response against a clearly scoped same-instruction reference.

This work does not introduce a new VLA architecture or action-chunk executor. Its contribution is an embodiment and

evaluation methodology: an action-level matched-input test prevents visually plausible driving from being mistaken for language use, while the paired closed-loop test determines whether the measured conditioning is large enough to affect the simulated vehicle trajectory.

D. Threats to Validity

The experiments use one simulator track, two matched starts, two lane intents, English instructions, one vehicle snapshot, and one fine-tuned checkpoint. Speed and destination slots are held semantically fixed, so no claim is made about their conditioning. The 32 closed-loop trials are sequential runs with policy seed 0, not independent training or simulator seeds; no significance claim is made. AB/BA balances instruction order, but SAME always precedes DIFFERENT within an invocation. Reset position is verified within 5 cm; yaw is logged but not an acceptance tolerance. The 20 matched observations come from three collection sessions in the training lineage, so they test conditioning interventions on fixed images, not visual or scene generalization. The symmetric test wording pair was selected after an exploratory offline probe; its factorial and closed-loop results characterize a known failure and are not an unbiased estimate of paraphrase success. The pointwise bootstrap treats the 20 matched observations as its resampling units; it does not model collection-session clustering, checkpoint-training uncertainty, or additional policy-sampling uncertainty. Rollouts use a 30-s wall-clock horizon while the control timer follows ROS/Gazebo simulation time; simulator real-time factor and elapsed simulation time were not stored. Consequently, the reported distance and wall time are not a standardized simulation-time speed benchmark.

An unadapted SmolVLA checkpoint has no compatible vehicle action normalizer, so we do not report a numerical pretrained-policy baseline and do not claim that fine-tuning improves a pre-existing driving score. Likewise, D_{chunk} and D_{shape} measure separation rather than route correctness, lane-boundary clearance, passenger comfort, or safety. No traffic, obstacles, weather, sensor faults, real vehicle, or functional-safety property is evaluated. The vehicle-shaped simulator uses differential-drive body dynamics; its visible front steering is not an Ackermann or real-automotive dynamics validation. Broader conclusions require multiple maps and starts, independently trained checkpoints, paraphrase families with graded wording differences, and goal- or lane-accuracy metrics in addition to the controlled conditioning tests used here.

V. CONCLUSION

We fine-tuned a compact SmolVLA to map a front image, an English driving instruction, and a checkpoint-matched state to a 30-step incremental SE(2) action chunk, then deployed it in a ROS 2–Gazebo loop. An offline matched-input 2^3 intervention localized the dominant loss of lane-conditioned action separation to replacing *inner/outer* with *innermost/outermost*; the conclusion persisted under the bridge-transform check. Across two starts and both instruction orders, Base closed-loop separation was 7.376 versus 0.177 m; every 0.2-m-resampled point after 15 m in all 16 Base rollouts was closer to

the requested reference. All 32 retained rollouts logged zero watchdog hits and zero action underrun. For all eight inner requests under the test wording, every corresponding point was instead closer to the outer reference. The result establishes a concrete language-to-motion path for the evaluated compact VLA while exposing a specific observed wording boundary rather than claiming broad natural-language generalization.

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